Live Coding Week 4 Mon

DATA 100, Spring 2026

```r
library(tidyverse)

# Load data
memi <- read_csv("data/memi.csv")
fmli <- read_csv("data/fmli.csv")

# Wrangling
fmli <- mutate(fmli, PRINEARN = as.integer(PRINEARN)) %>%
  left_join(memi, join_by("NEWID" == "NEWID", "PRINEARN" == "MEMBNO"))
```

- Let's find the average income and median retirement amount based on the sex of the principal earner.
  - Can I do two separate `summarize()`s?

```r
fmli %>%
  mutate(sex_prin_earner = case_when(
    SEX == 1 ~ "male",
    SEX == 2 ~ "female"
  )) %>%
  group_by(sex_prin_earner) %>%
  summarize(mean_inc = mean(FINCBTAX), med_retire = median(IRAX, na.rm = TRUE))

# Versus

fmli %>%
  mutate(sex_prin_earner = case_when(
    SEX == 1 ~ "male",
    SEX == 2 ~ "female"
  )) %>%
  drop_na(IRAX) %>%
```

```
  group_by(sex_prin_earner) %>%
  summarize(mean_inc = mean(FINCBTAX), med_retire = median(IRAX))
```

- Now find the average income and median retirement amount based on the sex of the principal earner for households where the principal earner is 40 or older.

```
fmli %>%
  mutate(sex_prin_earner = case_when(
    SEX == 1 ~ "male",
    SEX == 2 ~ "female"
  )) %>%
  filter(AGE >= 40) %>%
  group_by(sex_prin_earner) %>%
  summarize(mean_inc = mean(FINCBTAX), med_retire = median(IRAX, na.rm = TRUE))
```

- How do these summaries statistics vary by education level of household?

```
fmli %>%
  mutate(sex_prin_earner = case_when(
    SEX == 1 ~ "male",
    SEX == 2 ~ "female"
  )) %>%
  filter(AGE >= 40) %>%
  group_by(sex_prin_earner, HIGH_EDU) %>%
  summarize(mean_inc = mean(FINCBTAX), med_retire = median(IRAX, na.rm = TRUE))
```

- Now let's sort our table by average income.

```
fmli %>%
  mutate(sex_prin_earner = case_when(
    SEX == 1 ~ "male",
    SEX == 2 ~ "female"
  )) %>%
  filter(AGE >= 40) %>%
  group_by(sex_prin_earner, HIGH_EDU) %>%
  summarize(mean_inc = mean(FINCBTAX), med_retire = median(IRAX, na.rm = TRUE)) %>%
  arrange(desc(mean_inc))
```

- For rural places, what proportion of households have a female as the principal earner? What about for urban households?

```r
fmli %>%
  mutate(location = case_when(
    BLS_URBN == 1 ~ "Urban",
    BLS_URBN != 1 ~ "Rural"
  ),
  sex_prin_earner = case_when(
    SEX == 1 ~ "male",
    SEX == 2 ~ "female"
  )) %>%
  count(location, sex_prin_earner) %>%
  group_by(location) %>%
  mutate(prop = n/sum(n))
```